Python application programming interface (API)

import pyexocross as px

Input-file-driven command-line interface (CLI)

inp

python run.py

Run

Databases

par

pf OR txt

HITRAN

Convert data format

DB

ExoAtom

ExoMol

ExoMolHR

states trans

pf broad

csv pf

Input

PyExoCross

Download line list

www

Calculate

Physical and Spectroscopic Functionality

Thermodynamic and Radiative Quantities

States dependent

Partition functions

Specific heats

T dependent

Cooling functions

Radiative lifetimes

Oscillator strengths

Spectral Synthesis

Cross sections

Stick spectra

Plot

Transitions dependent

CPU

GPU

NVIDIA PyTorch CUDA

NVIDIA CuPy CUDA

PyTorch MPS

Binned Gaussian Gaussian

Binned Lorentzian Lorentzian

Binned Voigt

T and P dependent

SciPy Voigt

SciPy wofz Voigt

Humlíček Voigt

Thompson pseudo-Voigt

Kielkopf pseudo-Voigt

Olivero pseudo-Voigt

Liu-Lin pseudo-Voigt

Rocco pseudo-Voigt

Line Profile

Uncertainty filter

Threshold filter

Quantum number label filter

Quantum number value filter

Filter

Absorption

Emission

LTE

Two-temperature Treanor distribution

Custom vibrational density

Custom rovibrational population

Non-LTE

Output

log Log file

pf Partition function file

cp Specific heats file

cf Cooling functions file

states OR life Radiative lifetimes file

png os Oscillator strengths file

png stick Stick spectra file

png xsec Cross sections file

states trans broad HITRAN format / ExoMol format

par Other format / HITRAN format